Evolutionary Search

Genetic Algorithm I

FAST-NU, Lahore

GENETIC ALGORITHM

Introduction

A biologically inspired model of intelligence

The principles of *biological evolution* are applied to find solutions to difficult problems

The problems are not solved by reasoning logically about them; rather populations of competing candidate solutions are spawned and then evolved to become better solutions through a process patterned after *biological evolution*

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Introduction

Less worthy candidate solutions tend to die out, while those that show promise of solving a problem survive and reproduce by constructing new solutions out of their components

GA begin with a population of candidate problem solutions

Candidate solutions are evaluated according to their ability to solve problem instances: only the fittest survive and combine with each other to produce the next generation of possible solutions

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Introduction

Thus increasingly powerful solutions emerge in a Darwinian universe. Learning is viewed as a competition among a population of evolving candidate problem solutions

This method is heuristic in nature and it was introduced by John Holland in 1975

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Basic Algorithm

```
begin
  set time t = 0;
  initialise population P(t) = {x_1^t, x_2^t, …, x_n^t} of solutions;
  while the termination condition is not met do
    begin
      evaluate fitness of each member of P(t);
      select some members of P(t) for creating offspring;
      produce offspring by genetic operators;
      replace some members with the new offspring;
      set time t = t + 1;
    end
end
```

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Basic Attributes of Genetic Algorithm

1. **Genetic representation of candidate solutions (Chromosome)**
2. **Population size**
3. **Fitness (Evaluation) function**
4. **Genetic Operators**
5. **Selection technique/algorithm**
6. **Generation gap**
7. **Amount of elitism used**
8. **Number of duplicates allowed**

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Chromosome

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Representation of Solutions: The Chromosome

Gene*: A basic unit, which represents one characteristic of the individual. The value of each gene is called an *allele

***Chromosome*: A string of genes; it represents an individual i.e. a possible solution of a problem. Each chromosome represents a point in the search space**

***Population*: A collection of chromosomes**

An appropriate chromosome representation is important for the efficiency and complexity of the GA

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Representation of Solutions: The Chromosome

The classical representation scheme for chromosomes is binary vectors of fixed length. However, real valued chromosomes are also possible

In binary coded chromosomes, every gene has two alleles
In real coded chromosomes, a gene can be assigned any value from a domain of values

In the case of an n-dimensional search space, each chromosome consists of n variables with each variable encoded as a bit string

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Representation of Solutions: The Chromosome

Example: Cookies Problem

Two parameters sugar and flour (in kgs). The range for both is 0 to 9 kgs. Therefore a chromosome will comprise of two genes called sugar and flour

5	**1**

Chromosome # 01

2	**4**

Chromosome # 02

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Representation of Solutions: The Chromosome

Example: Expression satisfaction Problem

$$F = (\neg a \vee c) \wedge (\neg a \vee c \vee \neg e) \wedge (\neg b \vee c \vee d \vee \neg e) \wedge (a \vee \neg b \vee c) \wedge (\neg e \vee f)$$

Chromosome: Six binary genes a b c d e f e.g. 100111

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Representation of Solutions: The Chromosome

Example: Model Learning

Use GA to learn the concept *Yes Reaction* from the Food Allergy problem's data

Number	Restaurant	Meal	Day	Cost	Reaction
1	Sam's	breakfast	Friday	cheap	yes
2	Lobdell	lunch	Friday	expensive	no
3	Sam's	lunch	Saturday	cheap	yes
4	Sarah's	breakfast	Sunday	cheap	no
5	Sam's	breakfast	Sunday	expensive	no

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Example: Model Learning

A potential model of the data can be represented as a chromosome with the genetic representation:

Gene # 1 **Restaurant**	**Gene # 2** **Meal**	**Gene # 3** **Day**	**Gene # 4** **Cost**

The alleles of genes are:

Restaurant gene: Sam, Lobdell, Sarah
Meal gene: breakfast, lunch
Day gene: Friday, Saturday, Sunday
Cost gene: cheap, expensive

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Example

Find a number: 001010

You have guessed this binary number. If you write a program to find it, then there are $2^6 = 64$ possibilities

If you find it with the help of Genetic Algorithm, then the program gives a number and you tell its fitness

The fitness score is the number of correctly guessed bits

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Example

Find a number: 001010

Step 1. Chromosomes produced.

A) 010101 - 1
B) 111101 - 1
C) 011011 - 4 *
D) 101100 - 3 *

Best ones C & D

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Example

Find a number: 001010

	Mating	New Variants	Evaluation
C)	01:1011	01:1100 (E)	3
D)	10:1100	10:1011 (F)	4 *
C)	0110:11	0110:00 (G)	4 *
D)	1011:00	1011:11 (H)	3

Selection of F & G

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Example

Mating		New Variants	Evaluation
F)	1:01011	1:11000 (H)	3
G)	0:11000	0:01011 (I)	5 *
F)	101:011	101:000 (J)	4 *
G)	011:000	011:011 (K)	4

Selection of I and J

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Example

	Mating	New Variants	Evaluation
I)	0010:11	0010:00 (L)	5
J)	1010:00	1010:11 (M)	4
I)	00101:1	00101:0 (N)	6 (success) *
J)	10100:0	10100:1 (O)	3

In this game success was achieved after 16 questions, which is 4 times faster then checking all possible $2^6 = 64$ combination

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Representation of Solutions: The Chromosome

Rules Representation

Rules are often represented by bit strings (because they can be easily manipulated by genetic operators), but other numerical and symbolic representations are also possible

Set of if-then rules:

Specific sub-strings are allocated for encoding each rule pre-condition and post-condition

Example: Suppose we have an attribute "Outlook" which can take on values: *Sunny*, *Overcast* or *Rain*

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Representation of Solutions: The Chromosome

Rules Representation

We can represent it with 3 bits:

100 would mean the value *Sunny*,
010 would mean *Overcast* &
001 would mean *Rain*

110 would mean *Sunny* or *Overcast*
111 would mean that we don't care about its value

The pre-conditions & post-conditions of a rule are encoding by concatenating the individual representation of attributes

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Representation of Solutions: The Chromosome

Rules Representation

Example:

If (Outlook = Overcast or Rain) and Wind = strong
then PlayTennis = No

can be encoded as 0111001

Another rule

If Wind = Strong then PlayTennis = Yes

can be encoded as 1111010

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Representation of Solutions: The Chromosome

Rules Representation

An hypothesis comprising of both of these rules can be encoded as a chromosome 01110011111010

Note that even if an attribute does not appear in a rule, we reserve its place in the chromosome, so that we can have fixed length chromosomes

Note that 01110011111011
would mean, we do not care whether we play tennis or not

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Representation of Solutions: The Chromosome
Variable size chromosomes

Sometimes we need a variable size chromosome; e.g. to represent a set of rules
Example:
Suppose we are representing a set of rules by a chromosome

If	**a1 = T and**	**a2 = F**	**then c = T**
If	**a2 = T**		**then c = F**

If there is only the 1st rule in the set, we have 10 01 1
If the rule set has 1st & 2nd rules then we have 10 01 1 11 10 0

Chromosome Encoding for Simplified Time Table

- Courses = 16 (C1, C2, …..C16)
- Teachers = 8 (T1, T2, T3,……T8)
- Days = 5 (Mon, Tues, Wed, Thurs, Fri)
- Time-Slots = 5 (9-10, 10-11, 11-12, 12-1, 1-2)
- Classrooms = 4 (R1, R2, R3, R4)

- Chromosome encoding (Binary encoding)?
- Size of chromosome?

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2. Population Size

Number of individuals present and competing in an iteration (generation)

If the population size is too large, the processing time is high and the GA tends to take longer to converge upon a solution (because less fit members have to be selected to make up the required population)

If the population size is too small, the GA is in danger of premature convergence upon a sub-optimal solution (all chromosomes will soon have identical traits). This is primarily because there may not be enough diversity in the population to allow the GA to escape local optima

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Fitness Function

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Fitness Function

Also called "evaluation function"

It is used to determine the fitness of a chromosome

Creating a good fitness function is one of the challenging tasks of using GA

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Fitness Function

Example: Cookies Problem

Two parameters sugar and flour (in kgs). The range for both is 0 to 9 kgs. Therefore a chromosome will comprise of two genes called sugar and flour

5	**1**

2	**4**

The fitness function for a chromosome is the taste of the resulting cookies; values can be from 1 to 100

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Fitness Function

Example: Model Learning

Number	Restaurant	Meal	Day	Cost	Reaction
1	Sam's	breakfast	Friday	cheap	yes
2	Lobdell	lunch	Friday	expensive	no
3	Sam's	lunch	Saturday	cheap	yes
4	Sarah's	breakfast	Sunday	cheap	no
5	Sam's	breakfast	Sunday	expensive	no

Use GA to learn the concept *Yes Reaction* from the data

The *fitness function* can be the score of the classification accuracy of the rule-set over a set of provided training examples (number of training samples correctly classified by a chromosome)

Often other criteria may be included as well, such as the complexity of the rules or the size of the rule set

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Model Learning

Use GA to learn the concept *Yes Reaction* from the Food Allergy problem's data

Number	Restaurant	Meal	Day	Cost	Reaction
1	Sam's	breakfast	Friday	cheap	yes
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Chromosomes Encoding

A potential model of the data can be represented as a chromosome with the genetic representation:

Gene # 1	Gene # 2	Gene # 3	Gene # 4
Restaurant	Meal	Day	Cost

The alleles of genes are:

Restaurant gene: Sam, Lobdell, Sarah, X
Meal gene: breakfast, lunch, X
Day gene: Friday, Saturday, Sunday, X
Cost gene: cheap, expensive, X

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Population Size

Let the population size be 3 chromosomes/generation, because the instances are very few

Fitness Function

The *fitness function* can be the number of training samples correctly classified by a chromosome (model)

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Genetic Operators

Use the crossover operator, with randomly chosen single point crossover

Use mutation on one gene after every two iterations

Selection Algorithm

Proportionate selection

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Generation Gap

Variable, discard the 2 least fit chromosomes

Elitism

No elitism; discard the 2 least fit chromosomes

Duplicates

Duplicates are not allowed

Criteria for Termination

A chromosome correctly classifies all training samples

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1st Iteration

Suppose the following three chromosomes are created randomly:

Sam	**X**	**Sunday**	**cheap**
X	**breakfast**	**Sunday**	**X**
Sam	**breakfast**	**X**	**X**

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Number	Restaurant	Meal	Day	Cost	Reaction
1	Sam's	breakfast	Friday	cheap	yes
2	Lobdell	lunch	Friday	expensive	no
3	Sam's	lunch	Saturday	cheap	yes
4	Sarah's	breakfast	Sunday	cheap	no
5	Sam's	breakfast	Sunday	expensive	no

Sam	**X**	**Sunday**	**cheap**

X	**breakfast**	**Sunday**	**X**

Sam	**breakfast**	**X**	**X**

The fitness of 1st chromosome is 3
The fitness of 2nd chromosome is 1
The fitness of 3rd chromosome is 3

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1st Iteration

Suppose these two chromosomes are selected for crossover:

Sam	**X**	**Sunday**	**cheap**

Sam	**breakfast**	**X**	**X**

Suppose the crossover is after 2nd gene, and we get

Sam	**X**	**X**	**X**

Sam	**breakfast**	**Sunday**	**cheap**

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1st Iteration

Sam	**X**	**X**	**X**

Sam	**breakfast**	**Sunday**	**cheap**

The fitness of 1st chromosome is: 4
and that of 2nd is: 3

We reject the chromosome of fitness 1 and 3 and the rest of the three chromosomes (two old and one new) will go into the 2nd iteration

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Reproduction Operators

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Reproduction Operators

Genetic operators are applied to chromosomes that are selected to be parents, to create offspring

Basically of two types: Crossover and Mutation

Crossover operator create offspring by recombining the chromosomes of selected parents

Mutation is used to make small random changes to a chromosome in an effort to add diversity to the population

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Reproduction Operators: Crossover

Crossover operation takes two candidate solutions and divides them, swapping components to produce two new candidates

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Reproduction Operators: Crossover

Figure illustrates crossover on bit string patterns of length 8

The operator splits them and forms two children whose initial segment comes from one parent and whose tail comes from the other

Input Bit Strings

1 1 # 0 1 0 1 # **# 1 1 0 # 0 # 1**

Resulting Strings

1 1 # 0 # 0 # 1 **# 1 1 0 1 0 1 #**

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Reproduction Operators: Crossover

Two genes sugar and flour (in kgs)
Crossover operation on chromosomes

5	**1**
2	**4**

→

5	**4**
2	**1**

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Reproduction Operators: Crossover

The place of split in the candidate solution is an arbitrary choice. This split may be at any point in the solution

This splitting point may be randomly chosen or changed systematically during the solution process

Crossover can unite an individual that is doing well in one dimension with another individual that is doing well in the other dimension

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Reproduction Operators: Crossover

Two types: Single point crossover & Uniform crossover

Single type crossover

This operator takes two parents and randomly selects a single point between two genes to cut both chromosomes into two parts (this point is called cut point)

The first part of the first parent is combined with the second part of the second parent to create the first child

The first part of the second parent is combined with the second part of first parent to create the second child

1000010	**1000001**
1110001	**1110010**

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Reproduction Operators: Crossover

Uniform crossover

The value of each gene of an offspring's chromosome is randomly taken from either parent

This is equivalent to multiple point crossover

1000010

1110001 1010010

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Reproduction Operators: Crossover (Variable size chromosomes)

Suppose crossover points (1, 3) happen to be chosen for the 2nd parent

1st parent **10 01 1 11 10 0**
2nd parent **01 11 0 10 01 0**

The resulting two offspring would be

11 10 0

and

00 01 1 11 11 0 10 01 0

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Reproduction Operators: Mutation

Mutation is another important genetic operator

Mutation takes a single candidate and randomly changes some aspect (gene) of it

For example, mutation may randomly select a bit in the pattern and change it, switching a 1 to a 0 or to # (don't care)

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Reproduction Operators: Mutation

Mutation is important in that the initial population may exclude an essential component of a solution

For example, if no member of the initial population has a 1 in the first position, then crossovers cannot produce a child that could become a solution

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Reproduction Operators: Mutation

Each gene of each offspring is mutated with a given mutation rate p_μ (say 0.01)

It is hence possible that no gene may be mutated for many generations. On the other hand more than one gene may be mutated in the same generation (or even in the same chromosome)

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Reproduction Operators: Mutation

For real valued genes, the value is selected randomly from the alleles

If the rate is too low, new traits will appear too slowly in the population. If the rate is too high, each generation will be unrelated to the previous generation

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Reproduction Operators: Crossover (Important Similarities)

Suppose we have these chromosomes and their fitness values

A)	**01101**	**169**
B)	**11000**	**576**
C)	**01000**	**64**
D)	**10011**	**361**

If we analyze the strings which have high fitness value, we have observe that they both have a 1 in the 1st column, whereas the low fitness strings have a 0. In the second column we have a 1 in B and 0 in D, and 1 in the low fitness strings. We wonder that if 1 in the 2nd column is associated with low fitness, so we may improve the score of B by putting a 0 in place of 1 in the 2nd column

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References

Engelbrecht **Chapter 8 & 9**